

Research paper

Neck computed tomography angiography generation from computed tomography via Mamba U-shaped convolutional network-based diffusion with content and style conditions

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ABSTRACT

Computed tomography angiography (CTA) is a widely used imaging technique for diagnosing vascular diseases, particularly in detecting conditions such as aortic aneurysms and dissections. However, traditional CTA relies on iodinated contrast agents (ICAs), which pose risks to patients and present clinical challenges such as prolonged procedure times, high costs, and increased adverse reaction risks due to repeated use. To address these issues, this paper proposes a novel deep learning-based approach for generating CTA-like images from computed tomography (CT) scans, reducing dependence on ICAs and overcoming the limitations of conventional medical procedures. Specifically, we employ a conditional denoising diffusion probabilistic model and the Mamba U-shaped convolutional network as the artificial intelligence (AI) methods for implementation. To enhance image quality and precision, we introduce content and style conditioning modules to ensure anatomical and stylistic consistency with real CTA images. Content extractors capture key anatomical features from both CT and CTA images, and frequency-domain processing further enhances vascular structures. The style extractor ensures consistency in appearance with real CTA images. Experimental results demonstrate the superior performance of the proposed AI approach. On clinical private datasets of the neck and abdomen, the structural similarity index measure (SSIM) reached scores of 0.948 and 0.906, respectively, indicating the successful generation of high-quality images across multiple anatomical regions. This work represents a significant advancement toward non-invasive, safer, and higher-quality CTA image generation.

1. Introduction

Computed tomography angiography (CTA) is a non-invasive and effective vascular imaging technique widely used to diagnose lesions such as aortic aneurysms and dissections. However, it relies on iodinated contrast agents (ICAs), which pose risks for patients with iodine allergies, renal insufficiency, or multiple myeloma. Repeated ICA-based CTA is also costly and may increase the risk of adverse reactions.

Generating CTA-like images directly from non-contrast computed tomography (CT) offers several practical advantages: it may facilitate vascular assessment in patients with contraindications to contrast agents, reduce contrast exposure and associated costs during follow-up imaging of chronic vascular diseases, and support rapid vascular

visualization in emergency or screening scenarios. These considerations motivate the development of our CT-to-CTA synthesis framework.

The advent of generative adversarial networks (GANs) has substantially advanced medical image translation. Prior studies have leveraged architectures such as Pix2Pix (Isola et al., 2018), CycleGAN (Zhu et al., 2020), and Reg-GAN (Kong et al., 2021) for cross-modality translation, enabling effective image synthesis while alleviating the reliance on strictly paired datasets. These models aim to learn the underlying relationships between imaging modalities and generate images that resemble real anatomical structures. Accordingly, GAN-based approaches have also been explored for CT-to-CTA synthesis (Lyu et al., 2023). Despite these advances, GAN-based models still suffer from several

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inherent limitations, including unstable training dynamics, susceptibility to generating anatomically implausible structures, and increased training complexity (Y. Li et al., 2023).

In recent years, denoising diffusion probabilistic models (DDPMs) (Ho et al., 2020) have emerged as a powerful framework for multi-domain image synthesis by modeling a stochastic forward and reverse diffusion process. However, purely unsupervised diffusion models often show limited capability in cross-modal generation and task-specific adaptation. To address these issues, conditional diffusion models, such as the conditional denoising diffusion probabilistic model (CDDPM) (Choi et al., 2021) and the denoising diffusion model with multiple tasks (DDMM) (Ye and Xu, 2024), introduce external conditions or multi-task learning mechanisms to guide the denoising process toward target distributions. Despite these advances, diffusion-based methods still face challenges in medical imaging applications, including limited anatomical detail fidelity, data scarcity, and high computational cost.

Motivated by these limitations, this paper addresses the challenge of high-detail medical image generation by introducing a content-style conditioned diffusion framework, termed computed tomography angiography diffusion (CTA-Diff), for CT-to-CTA synthesis. To ensure anatomical consistency with both the source CT and the target CTA, the proposed framework incorporates dual content encoders, which extract complementary structural information from CT and CTA images, respectively. In addition, a style encoder is employed to capture multi-level CTA characteristics and provide global stylistic guidance during the diffusion process, promoting consistency in both anatomical structure and visual appearance. To improve computational efficiency, CTA-Diff integrates a Mamba U-shaped convolutional network (Mamba-Unet) (Wang et al., 2024) as the noise prediction module. Mamba, a state space model-based sequence architecture, offers linear computational complexity and has demonstrated strong efficiency in modeling long-range dependencies across multiple domains. These properties make it well suited for diffusion-based medical image synthesis, where both structural fidelity and scalability are critical. Experimental results on neck and abdominal CT-CTA datasets demonstrate that the proposed CTA-Diff model is capable of generating high-quality CTA-like images with accurate anatomical structures and favorable generalization performance. Ablation studies further validate the effectiveness of each proposed component.

The main contributions of this work are as follows:

- We introduce a novel diffusion model built upon the Mamba-Unet architecture, effectively integrating rich cues to achieve high-quality CTA image generation with accelerated inference and superior detail retention.
- We designed two modality-specific encoders for CT and CTA, integrated with a discrete cosine transform (DCT)-based algorithm to emphasize critical regions and ensure anatomical consistency between generated and real CTA images.
- A CTA-style encoder is introduced to extract and normalize multi-level feature styles, guiding the generation of images with distinct CTA characteristics.
- A style contrast loss is introduced to enforce consistency between the generated image and the distinctive CTA style by comparing style vectors extracted from generated and real CTA images in the feature space.

2. Related works

2.1. Image-to-image translation

The image-to-image (I2I) translation task aims to enable transformations between images in source and target domains. Early studies relied heavily on GANs for I2I translation, laying the foundation for

the field (Goodfellow et al., 2014). In recent years, GAN-based models have continued to dominate this domain. Dalmaz et al. (2022) enhanced medical image generation by incorporating Transformers' context sensitivity into GAN architectures. To address modality incompleteness, Zhang et al. (2025b) proposed a unified multimodal GAN approach, leveraging both modality-invariant and modality-specific information. Bharti et al. (2024) introduced a multi-objective cyclical GAN with evolutionary optimization to improve generation quality while preserving key features. Atli et al. (2024) incorporated state-space models and channel-mixing Mamba blocks into convolutional backbones, enhancing cross-spatial context learning for high-fidelity image generation.

However, due to the inherent instability of GAN training, diffusion models have recently emerged as a promising alternative. Saharia et al. (2022) introduced a smooth and efficient image-to-image diffusion model that outperforms traditional GANs across various tasks. Zhan et al. (2024) further proposed a cross-guided alignment diffusion method to align multiple medical modalities using limited paired data, extracting clinically relevant invariant features. Yi et al. (2024) developed a multi-modal fusion paradigm by integrating prior knowledge with diffusion models, achieving high-quality fused images. Luo et al. (2024) employed a perceptual priority weighting scheme and a pre-trained classifier, producing visually realistic outputs. Recent diffusion models have also incorporated frequency information to enhance detail control and fidelity. Xing et al. (2024) learned target modality distributions by randomly masking image patches and reconstructing them, then decoupled noise to synthesize the target modality via fusion. Y. Li et al. (2023) guided the diffusion process using frequency-domain filtering, preserving structural information and enabling zero-shot generation without source domain data. Zhang et al. (2025a) introduced a texture-preserving diffusion model with an adaptive wavelet-based high-frequency optimization module, enhancing feature extraction and denoising while maintaining structural consistency.

These studies highlight the increasing prominence of diffusion models in medical image generation and their potential as a core framework for I2I translation. CT-to-CTA translation is one of the most important applications in vascular imaging. To address this challenge, Lyu et al. (2023) proposed a GAN-based method that improved generation quality. Building on this, we propose a style-conditioned diffusion framework specifically designed for CT-to-CTA translation, achieving higher generation quality while preserving fine vascular structures.

2.2. Diffusion model

In recent years, diffusion models have achieved remarkable progress in image generation by modeling data distributions via a forward diffusion process and reconstructing noise-free images through a reverse denoising process. A key milestone is the DDPM (Ye and Xu, 2024), which optimizes the variational bound by combining denoising score matching with probabilistic diffusion, enabling high-quality image synthesis. Song et al. (2020) proposed an implicit diffusion process to improve generation efficiency, while Rombach et al. (2022) introduced latent diffusion with variational autoencoders, reducing input dimensionality and enhancing modeling efficiency.

With the emergence of Transformers, Peebles and Xie (2023) employed them as noise predictors to improve computational performance. Yang et al. (2022) integrated vision Transformers (ViT) into DDPM via the GenViT model, bridging discriminative and generative paradigms, and Bao et al. (2023b) designed a general-purpose ViT-based architecture for large-scale cross-modal generation. Hatamizadeh et al. (2024) proposed a time-dependent self-attention mechanism for adaptive attention during denoising, yielding high-fidelity results.

More recently, Mamba has further advanced diffusion models. Hu et al. (2024) proposed the parameter-free Zigzag Mamba to enhance speed and memory efficiency, while Teng et al. (2024) integrated Mamba into diffusion backbones for efficient high-resolution image

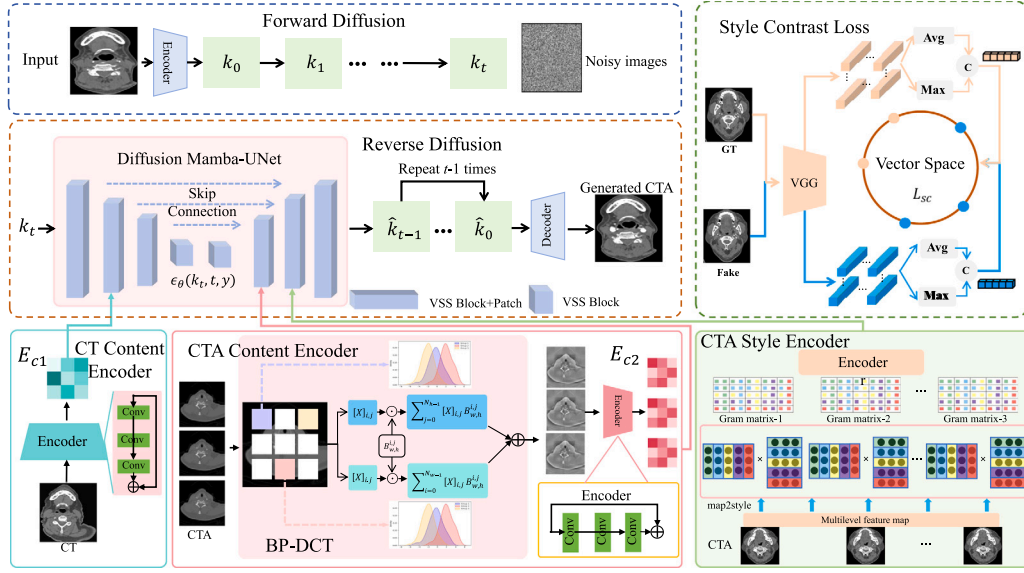


Fig. 1. The overall architecture of CTA-Diff consists of a Mamba-UNet-based conditional diffusion model, incorporating a CT content encoder, a CTA content encoder, and a CTA style encoder. A style contrast loss is introduced to emphasize CTA-specific styles and provide effective guidance for the diffusion process. Both the generated and ground-truth CTA images are mapped into a style embedding vector space via VGG features to quantify their stylistic discrepancy.

generation. Distinct from prior work, this study employs Mamba-UNet as the denoising backbone, leveraging its ability to capture long-range dependencies and fine-grained structures. This design facilitates accurate contextual modeling and detail preservation, leading to superior performance in medical image generation tasks.

3. Methodology

The overall framework of CTA-Diff is illustrated in Fig. 1. CTA-Diff operates as a latent diffusion model: the input CT image is encoded into a low-dimensional latent space, enabling the generation of paired CTA images under style and content guidance. In the forward diffusion process, Gaussian noise is gradually added to the latent representation until it becomes pure noise. The reverse process then denoises this representation to reconstruct the corresponding CTA image, guided by three core components: (i) Mamba-UNet, serving as the denoising backbone and modeling the noise distribution during generation; (ii) Modality-specific content encoders for CT and CTA, extracting structural features to ensure anatomical consistency; (iii) CTA style conditioning module, capturing and injecting multi-scale style cues from real CTA images to guide generation toward authentic vascular appearance.

Compared with existing CT-to-CTA and diffusion-based image-to-image translation methods, CTA-Diff introduces several methodological novelties. First, instead of a conventional UNet or Transformer backbone, we adopt a Mamba-UNet state-space model as the noise predictor in a conditional diffusion framework, improving global context modeling and high-resolution inference. Second, we design dual modality-specific content encoders, including a CTA encoder enhanced by band-pass discrete cosine transform (BP-DCT), to align CT-CTA anatomical content in the latent space and emphasize vascular structures. Third, we introduce a CTA style conditioning with a contrastive style loss in a learned style-vector space, enforcing structural fidelity and multi-slice visual style consistency with real CTA. These components jointly distinguish CTA-Diff from prior GAN- and diffusion-based CT-to-CTA methods and form the core of our framework.

3.1. Diffusion model based on Mamba-UNet

The CT-to-CTA image conversion task aims to transform CT images into corresponding CTA images, ensuring that the generated CTA

closely resembles the true CTA scan. Our primary focus is on generating high-quality images of the carotid and vertebral arteries, which are critical for clinical applications. The denoising predictor is built upon the classical denoising diffusion implicit models (DDIM) framework. While DDIM typically employs UNet or Transformer as the noise predictor, we enhance this by integrating Mamba-UNet, which improves the model's ability to capture global context and intricate details from CT slices. Mamba-UNet takes grayscale two-dimensional (2D) images of size $H \times W \times 1$ as input, which are first segmented into patches similar to the methodology used in vision Mamba (VMamba) (Liu et al., 2024). We adopt a U-shaped Mamba architecture as the denoising backbone, referred to as Diffusion Mamba-UNet (Fig. 2). The network follows an encoder-decoder design with skip connections. In the encoder, after patch partition and linear embedding, a sequence of visual state space (VSS) blocks and patch-merging layers progressively reduces the spatial resolution while increasing the channel width. The decoder mirrors this structure with patch-expanding layers and additional VSS blocks, and the output is finally projected back to the image space. Compared with the original Mamba-UNet designed for segmentation, our Diffusion Mamba-UNet is modified in two aspects. First, the segmentation head is replaced with a single-channel regression head, enabling the network to predict continuous CTA intensities instead of class logits. Second, the depth and channel configuration are adjusted to accommodate 512×512 CT slices and serve as a diffusion denoiser, while the core VSS blocks and patch merging/expanding mechanisms remain unchanged. CTA-Diff operates in a two-dimensional setting. The Mamba-UNet denoising backbone takes a single axial CT slice $x_{CT} \in \mathbb{R}^{H \times W}$ as input and predicts the corresponding axial CTA slice $\hat{x}_{CTA} \in \mathbb{R}^{H \times W}$, establishing a one-to-one mapping between input and output slices. Patient-level three-dimensional (3D) CTA volumes are then reconstructed by stacking all generated axial slices according to the original digital imaging and communications in medicine (DICOM) order and spacing.

3.1.1. VSS block

The VSS block is primarily based on Visual Mamba (Zhu et al., 2024). After layer normalization, the input is split into two branches: one branch passes through a linear layer, while the other undergoes a linear layer followed by depth-wise separable convolution. The features are then further processed by a 2D selective scan module (SS2D) (Liu et al., 2024). Finally, the extracted features are normalized again and combined with a residual connection to produce the block's output.

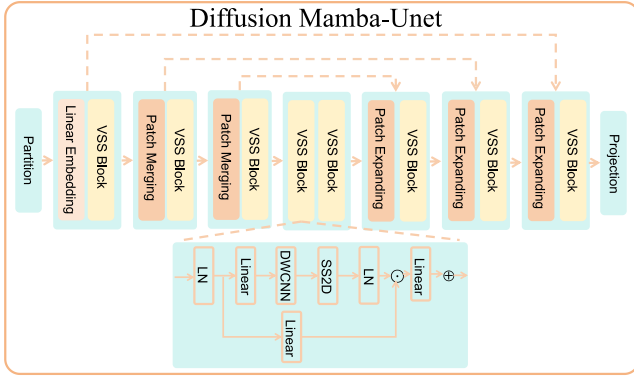


Fig. 2. Structure of the proposed Diffusion Mamba-UNet backbone: The network adopts a U-shaped encoder–decoder architecture with VSS blocks and patch merging/expanding layers. A regression projection head replaces the original segmentation head, enabling the network to serve as the denoising backbone for diffusion-based CT-to-CTA synthesis.

3.1.2. Patch merging and patch expanding

Patch merging is a more sophisticated operation than traditional pooling, as it preserves all information without loss. Each downsampling step reduces the spatial resolution by a factor of two by selecting elements at intervals of two along both row and column directions to form new patches. These patches are concatenated into a single tensor and unfolded, increasing the channel dimension fourfold. A fully connected layer then reduces the channel dimension to twice its original size.

Our patch merging extends the original design by incorporating scan switching. While conventional patch merging creates a unidirectional receptive field, our method achieves a global receptive field by scanning patches in multiple directions: in the first block, a row-wise scan is performed from left to right; in the second block, the sequence is reversed. Subsequent blocks scan columns in both forward and reverse directions. After completing all scan patterns, the process loops back for the next block.

The patch expansion layer performs upsampling on the deep features and halves the feature dimensions. Conceptually, it mirrors the patch merging operation in reverse, ensuring a balanced reconstruction of image features.

Formally, let the input feature map to a patch merging layer be

$$X \in \mathbb{R}^{H \times W \times C}, \quad (1)$$

where H and W denote the spatial resolution, and C is the number of channels. Each downsampling step reduces both height and width by a factor of 2. For each non-overlapping 2×2 window, the four tokens are concatenated along the channel dimension:

$$Y_{i,j} = [X_{2i,2j}, X_{2i+1,2j}, X_{2i,2j+1}, X_{2i+1,2j+1}] \in \mathbb{R}^{4C}, \quad (2)$$

for $i = 0, \dots, H/2 - 1$ and $j = 0, \dots, W/2 - 1$. This yields an intermediate tensor

$$Y \in \mathbb{R}^{(H/2) \times (W/2) \times 4C}. \quad (3)$$

A fully connected layer is then applied independently at each spatial location to reduce the channel dimension from $4C$ to $2C$.

$$Z_{i,j} = W_m Y_{i,j} + b_m, \quad (4)$$

where $W_m \in \mathbb{R}^{2C \times 4C}$ and $b_m \in \mathbb{R}^{2C}$, producing the downsampled feature map

$$Z \in \mathbb{R}^{(H/2) \times (W/2) \times 2C}. \quad (5)$$

Thus, patch merging performs spatial downsampling (stride 2), aggregates local tokens to enlarge the receptive field, and enhances channel-wise representational capacity.

Following VMamba, we reshape Z into a 1D token sequence for the VSS blocks. Rather than a single unidirectional scan, we adopt a scan-switching strategy: the first block scans rows left-to-right, the second rows right-to-left, the third columns top-to-bottom, and the fourth columns bottom-to-top, then the pattern repeats. This generalizes VMamba's original scheme, allowing the model to capture long-range dependencies along multiple directions while preserving spatial structure.

The patch expansion layer performs the inverse operation of patch merging. Given

$$U \in \mathbb{R}^{(H/2) \times (W/2) \times 2C}, \quad (6)$$

a linear layer first expands the channel dimension from $2C$ to $4C$. The resulting features are then rearranged to restore the original spatial resolution,

$$\tilde{X} \in \mathbb{R}^{H \times W \times C}, \quad (7)$$

by redistributing the $4C$ channels into the four spatial positions of each 2×2 neighborhood. This enables a learned upsampling that is strictly aligned with the preceding patch merging, allowing the decoder to recover fine-grained spatial details from the compressed representations.

3.2. Content encoders adapted to different modalities

In conditional diffusion models, given an original image x_c and a reference image x_r , the goal is to train a model that generates an output image that preserves the content consistency of x_c . While CT and CTA images share the same underlying content, extracting relevant information from CTA images is typically more straightforward. To effectively capture content from both modalities, we design separate content extractors for CT and CTA images. These extractors independently extract content features, which are then used as conditional inputs for the denoising network.

3.2.1. Content encoder for CT images

For CT images, though less prominent and harder to extract than in CTA images, texture features still contain all anatomical information necessary. To ensure anatomical consistency in the generated CTA image, feature extraction from CT is essential. Convolutional neural networks (CNNs) are commonly used for this purpose, mapping high-dimensional CT images to lower-dimensional feature spaces through multiple convolutional layers. In this work, we adopt ResNet34 as the CT content extractor. Since the denoising network is based on Mamba-UNet, we apply padding and flattening to align the extracted CT feature maps with its input requirements for content conditioning.

3.2.2. Content encoder for CTA images

CTA and CT images exhibit significant differences. CTA images emphasize vascular structures, facilitating the extraction of blood vessel features. However, conventional deep learning models such as ResNet are not specifically designed for vascular feature extraction, which makes it challenging to capture fine vascular details accurately. To overcome this limitation, we apply a series of preprocessing steps to CTA images, enhancing vascular structures for more effective feature extraction.

Recent studies (Z. Li et al., 2023; Xu et al., 2020) have demonstrated the effectiveness of using the DCT spectrum for image feature learning. DCT exhibits strong energy compaction, preserving essential low-frequency components while suppressing irrelevant high-frequency noise—an advantage in medical imaging tasks. However, the contribution of different frequency bands varies. High-frequency components enhance edge details but may obscure critical vascular structures due to the loss of low-frequency content, whereas low-frequency components maintain anatomical consistency but fail to highlight fine

vascular features. To address this, we propose a band-pass DCT (BP-DCT) strategy that selectively retains mid-frequency components – approximately corresponding to 5–50 Hz – while discarding high-frequency noise and low-frequency background. This tailored spectral filtering enhances vascular feature representation in CTA images while effectively suppressing irrelevant CT content.

The CTA content encoder follows a structure similar to the CT content encoder, with the key difference being the integration of a BP-DCT block for spectral filtering. The CTA image is first processed by this band-pass filter to enhance vascular features, and the resulting image is then encoded to extract content representations. These features are subsequently fed into the denoising network for content conditioning. While fundamentally similar to the CT content extraction process, this design is optimized to capture fine vascular information in CTA images.

3.3. Style encoder for CTA images

After extracting multi-level VGG features from CTA images and computing their Gram matrices, these features are projected into a low-dimensional style embedding $v \in \mathbb{R}^d$. The collection of such embeddings defines a style vector space, referred to as the “vector space” in Fig. 1. Each CTA image (real or generated) is thereby represented by a compact style vector in this space, which is then used to measure style similarity. Image style transfer has attracted significant attention in recent years, aiming to impose the artistic style of a reference image onto the content of a target image while preserving its structure. In the context of CT and CTA images, CTA emphasizes vascular features derived from CT, while both modalities share the same underlying anatomy. From this perspective, generating CTA from CT can be interpreted as a specialized form of style transfer, where the “style” corresponds to vascular appearance and contrast.

As mentioned, content encoders extract features from both CT and CTA images to ensure anatomical alignment of the generated CTA with the original CT. In addition to this content guidance, generating a CTA image also requires style guidance specific to CTA characteristics.

In image style transfer tasks, the VGG network (Simonyan and Zisserman, 2015) is widely used for style encoding due to its strong feature representation and simple architecture. It captures multi-level texture features across different abstraction levels: shallow layers focus on low-level details like edges and corners, middle layers capture complex textures and patterns, and deeper layers extract high-level structural and semantic information. Typically, features from the middle and deep layers are used to represent rich style textures. The style information is encoded via the Gram matrix, computed from the inner products of the feature maps, which quantifies the correlations between channels and thus captures the distribution of textures, patterns, and other style elements. Given a convolutional layer’s feature map F , the Gram matrix G is computed as:

$$G_{ij}^l = \sum_k F_{ik}^l F_{jk}^l, \quad (8)$$

Here, F_{ik}^l and F_{jk}^l denote the k th features of the i th and j th channels in the feature map at layer l , and G_{ij}^l represents the correlation between channels i and j . By minimizing the difference between the Gram matrices of the generated and style images across multiple layers, the network adjusts the texture of the generated image to match the target style.

In the proposed architecture, the CTA image is fed into a feature extractor with an encoder–decoder structure to progressively capture multi-level features. The Gram matrix G is then computed at each level to represent complex style features. To balance style characteristics across scales, informative content is maximized while redundancy is suppressed, yielding the adjusted style representation $\tilde{G}$, which serves as the encoded style condition. The overall style encoding procedure E_s is thus formulated as:

$$\tilde{G} = F(G \mid \mu, v, o), \quad \text{with } \mu, v, o \in \mathbb{R}^c. \quad (9)$$

The embedded input is adjusted by embedding weight μ . Self-activation is controlled by learnable weights v and bias o . Let G_c represent the channels in G , where $G_c = [G_c^{(m,n)}]_{(h \times w)} \in \mathbb{R}^{(h \times w)}$. Let the embedding weight vector $\mu = [\mu_1, \mu_2, \dots, \mu_c]$, and μ can be adjusted to P (μ_c as an example, we can associate the embedded weight with a channel in G and turn it into P_c):

$$P_c = \mu_c \|G_c\|_2 = \mu_c \left[\left(\sum_{m=1}^h \sum_{n=1}^w (G_c^{(m,n)})^2 \right) + \beta \right]^{\frac{1}{2}}. \quad (10)$$

Here, β is a small constant introduced to prevent numerical instability when differentiation at zero, and $\|\cdot\|_2$ denotes the L_2 norm. This normalization step facilitates stable competition among channels, promoting more balanced feature activation. The set of operations formed as $P = \{P_c\}_{c=1}^Q$. We apply L_2 normalization across channels:

$$\hat{P}_c = \frac{\sqrt{Q} P_c}{\|P\|_2} = \frac{\sqrt{Q} P_c}{\left(\sum_{c=1}^Q P_c^2 + \beta \right)^{1/2}}, \quad (11)$$

where Q denotes the scale. An activation mechanism updates each channel, enabling the diffusion model to fully leverage style features at different scales during training, thereby improving synthesis performance. Given learnable weights $v = [v_1, \dots, v_c]$ and biases $o = [o_1, \dots, o_c]$, we compute:

$$\tilde{G}_c = G_c [1 + S(v_c \hat{G}_c + o_c)]. \quad (12)$$

This equation provides the new representation of each compressed style condition after automatic weighting. Here, $S(\cdot)$ denotes the Sigmoid activation function.

3.4. CTA style contrast loss

We formulate the generation of CTA images from CT images as a style transfer problem. In addition to using a style extractor to capture CTA characteristics as a conditioning factor, we introduce a contrastive loss to enhance alignment between generated and real CTA images, ensuring consistency at both global and local levels. Specifically, the architecture of the proposed style contrast loss (SCL), which incorporates a style extractor, is illustrated in Fig. 1. Building on the work of Zhang et al. (2022a) and others, we employ a VGG network to process both real and generated CTA images through a style extractor. To effectively capture both global and local style features, we extract features from N layers of the VGG network, represented as $F_v = (f_v^0, f_v^1, f_v^2, \dots, f_v^N)$, which serve as inputs for style projection. To obtain distinct global channel features, we apply both average pooling and max pooling, then concatenate the results channel-wise to form the feature set $F_g = (f_g^0, f_g^1, f_g^2, \dots, f_g^N)$. After a series of linear projections, we derive the style vector $V = \{v^0, v^1, v^2, \dots, v^N\}$, which acts as a supervisory signal for the diffusion model, guiding it to replicate the desired style. To further enforce style consistency, we adopt a style contrast strategy, supervising the generated sample $\hat{x}_0$ to align with the target style. This is achieved using the style contrast loss $\mathcal{L}_{sc}$, which leverages the style vectors V_s^1 and V_s^2 , already embedded in the style vector space. In this style vector space, real and generated CTA images are mapped to vectors that encode their global appearance. The style contrast loss operates on pairs of style vectors, encouraging matched real-generated pairs to be close while pushing mismatched pairs apart. This guides the diffusion model to produce images whose visual style aligns with real CTA in the learned vector space. The formulation of $\mathcal{L}_{sc}$ is as follows:

$$\mathcal{L}_{sc} = - \sum_{s=0}^{N-1} \log \left(\frac{\exp(V_s^1 \cdot V_s^2 / \tau)}{\exp(V_s^1 \cdot V_s^2 / \tau) + \exp(V_s^1 \cdot V_s^2 / \tau)} \right), \quad (13)$$

where N denotes the batch size, τ is a temperature parameter that scales the activation, and s indexes the samples. In addition to the style contrast loss, we employ a mean squared error (MSE) loss as a key objective. In our diffusion model, the final CTA images are generated

by iteratively removing the predicted noise, with the MSE loss at each step defined as:

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{j=1}^N (out_j^i - e_{gt}^i)^2, \quad (14)$$

where out_j represents the predicted noise for the j th sample, and e refers to the noise added to the j th sample at the corresponding step in the forward diffusion process. The total loss function is formulated as:

$$\mathcal{L}_{total} = \mathcal{L}_{MSE} + \alpha_{sc} \cdot \mathcal{L}_{sc}, \quad (15)$$

where α_{sc} is a hyperparameter that adjusts the impact of the style contrast loss in the overall objective to guide selection of the optimal generation model. We follow the standard noise-prediction loss without noise-level scaling, leaving a detailed exploration of diffusion models (EDM)-style noise-dependent weighting for future work.

In this study, the training objective combines the standard MSE noise-prediction loss with the proposed style contrast loss. The structural similarity index measure (SSIM) is used solely for test-time evaluation; it is not included as an auxiliary loss to maintain stable and simple optimization of the diffusion backbone on our limited dataset. Investigating the effect of SSIM or other perceptual losses on CT-to-CTA diffusion training is left for future work.

Notably, CTA-Diff is trained with the conventional MSE noise-prediction loss, mathematically equivalent to a score-matching objective in diffusion models. The style contrast loss acts as an auxiliary objective in a separate style-embedding space rather than replacing the diffusion loss. In this space, each real or synthesized CTA is mapped to a global style vector, and the contrastive formulation encourages the generated CTA to be stylistically closer to its paired real CTA than to other subjects. Such pairwise style discrimination is naturally expressed via contrastive learning and is difficult to enforce through score-matching or pixel-level losses, which mainly supervise local intensity consistency. Empirically, combining the diffusion loss with the style contrast loss produces more coherent CTA appearances across slices and subjects, while preserving the core score-matching behavior of the diffusion model.

4. Experiments and results

4.1. Experiments setup

4.1.1. Dataset

Dataset overview. Two independent clinical CT-CTA datasets were used: a neck (carotid) cohort and an abdominal cohort, each comprising 25 patients (50 unique patients in total). All subjects were routine clinical patients at Gongli Hospital of Shanghai Pudong New Area (Gongli Hospital) undergoing non-contrast CT and contrast-enhanced CTA for suspected or known vascular diseases; no healthy volunteers were recruited. CT and CTA scans were acquired during the same imaging session to minimize temporal differences. Data were collected between July 2017 and June 2024, stored in DICOM format, and acquired on Siemens CT scanners.

Acquisition protocols. For each subject, non-contrast CT and contrast-enhanced CTA scans were strictly paired and acquired consecutively in a single session on the same Siemens CT scanner, with identical positioning, field of view, and acquisition parameters. Both neck and abdominal datasets consist of contiguous slices with 0.75 mm thickness, where the slice spacing equals the slice thickness (i.e., zero inter-slice gap), and a 512×512 in-plane resolution. CTA scans used iodinated contrast (370 mg/ml). Neck volumes typically comprise 300–400 slices, while abdominal volumes follow the same acquisition protocol, ensuring consistency in key scanning parameters apart from anatomical coverage. All CT-CTA pairs are temporally matched and inherently aligned, acquired from the same CT scanner in the same session, with no additional registration required, as the CT-CTA pairs

exhibited negligible motion or misalignment upon inspection. This intrinsic alignment ensures that each CT slice corresponds directly to a matching CTA slice, enabling reliable supervised learning for CT-to-CTA synthesis.

Train-validation-test splits. Each cohort was treated independently, with subject-level splits to prevent data leakage. For both cohorts, 17 subjects were used for training, 3 for validation, and 5 for testing ($\sim 0.7/0.1/0.2$ ratio). Neck CTA slices were manually segmented by an experienced radiologist for reference, while CTA-Diff and all baseline methods were trained on full CTA images. Preprocessing – including DICOM loading, resampling to 512×512 , and intensity clipping/normalization – was applied consistently across all methods. Hyperparameters were tuned solely on the validation set, and quantitative results were reported on the held-out test set, ensuring fair comparisons.

Summary. The neck and abdominal datasets are retrospective, single-center collections, not previously publicly released. Treating the cohorts independently and standardizing acquisition, preprocessing, splitting, and evaluation guarantees that reported performance differences reflect model behavior rather than variations in data handling.

4.1.2. Implementation details

Data preprocessing. The paired CT-CTA dataset comprises 50 clinical subjects, allowing efficient model training. Images were resampled to 512×512 to preserve anatomical structures.

Manual annotation. For the neck cohort, carotid arteries were manually segmented on each axial CTA slice by an experienced radiologist, producing voxel-level lumen masks that serve as reference standards for vessel boundaries. These masks were used for 3D vascular reconstruction and qualitative assessment, while CTA-Diff was trained in an image-to-image manner using the full CTA images as supervision.

Data splitting and patient-based design. Data splitting was performed at the subject level to prevent leakage: all slices from a given patient were assigned exclusively to the training, validation, or test set. During training, 2D slices were randomly sampled from the training subjects to form mini-batches. During evaluation, slice-wise image-quality metrics were averaged across all slices of each subject to obtain volume-level scores.

Network training and implementation. Networks were implemented in PyTorch and trained on an NVIDIA GeForce RTX 4090D GPU. A batch size of 4 was used, and optimization was performed with AdamW. Following previous works (Bao et al., 2023b), we adopted the DDIM scheduler, a pre-trained image autoencoder (Kingma et al., 2013), and a diffusion probabilistic model (DPM) solver (Lu et al., 2022). The learning rate was set to 2×10^{-4} to ensure stable convergence.

4.1.3. Hyperparameter selection in loss function

Experiments were conducted to determine the optimal value of the key parameter α_{sc} for both the neck and abdominal datasets, with results shown in Fig. 3. For the neck dataset, the model achieved a peak signal-to-noise ratio (PSNR) of 38.88 dB and an SSIM of 0.94 when $\alpha_{sc} = 0.1$. In contrast, setting α_{sc} to 0.001 led to a marked performance drop, with PSNR decreasing to 30.08 dB (a decline of 8.8 dB) and SSIM decreasing by 0.13. A similar trend was observed in the abdominal dataset, with slightly lower overall metrics: the highest PSNR was 37.92 dB, which was 10.45 dB higher than the result obtained with $\alpha_{sc} = 20$, while SSIM reached 0.90. These results suggest that setting $\alpha_{sc} = 0.1$ yields optimal generation performance across both anatomical regions.

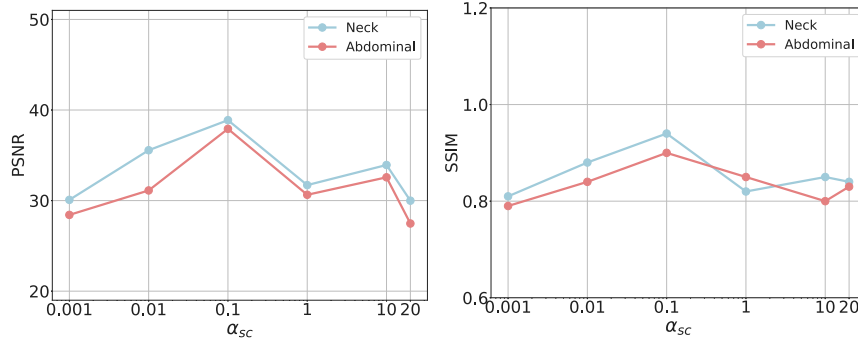


Fig. 3. Impact of hyperparameter α_{sc} on network performance.

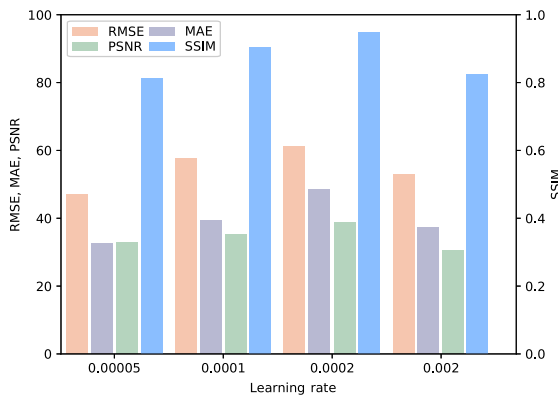


Fig. 4. Effect of learning rate on CTA generation.

4.1.4. Effect of learning rate on CTA generation

The CTA-Diff was systematically evaluated under four distinct learning rates (0.00005, 0.0001, 0.0002, and 0.002), selected based on preliminary tests indicating their notable influence on learning dynamics. As shown in Fig. 4, the model achieved the best performance at a learning rate of 0.0002, attaining an SSIM of 0.948 and a PSNR of 38.87, indicative of effective feature representation and high-fidelity generation. In contrast, a higher rate of 0.002 resulted in significant performance degradation (SSIM: 0.823; PSNR: 30.63), likely due to training instability and convergence failure. Similarly, an excessively low rate of 0.00005 also hindered performance (SSIM: 0.813; PSNR: 32.88), reflecting a trade-off between convergence speed and stability. Based on these observations, a learning rate of 0.0002 was selected as the optimal setting for CTA-Diff and was employed in all subsequent experiments.

4.2. Comparison with the state-of-the-art methods

To thoroughly evaluate CTA-Diff, we compared it with some advanced medical image-to-image translation techniques, including frequency-guided diffusion model (FGDM) (Y. Li et al., 2023), generative adversarial network-based noncontrast CT angiography (CTA-GAN) (Lyu et al., 2023), image-to-image Mamba (I2I-Mamba) (Atli et al., 2024), cross-conditioned diffusion model (CDM) (Xing et al., 2024), medical multi-modal generative framework (MedM2G) (Zhan et al., 2024), multi-modality image fusion via Diffusion model with fusion knowledge prior (Diff-IF) (Yi et al., 2024), and texture-preserving diffusion model (TPDM) (Zhang et al., 2025a).

For quantitative evaluation, four image quality metrics were used: PSNR, SSIM, normalized mean absolute error (NMAE), and mean absolute error (MAE, used in the ablation study). The network outputs

two-dimensional axial CTA slices. For each test subject, metrics were first computed slice-wise between each predicted CTA slice and its corresponding ground-truth CTA slice, and then averaged across all slices within the volume to obtain a patient-level value. Reported values in all tables represent the mean $\pm$ standard deviation of these patient-level scores across all test subjects for the neck and abdominal cohorts, respectively.

4.2.1. Qualitative analysis

For qualitative evaluation, we conducted visual comparisons between CTA images generated by different methods and our proposed CTA-Diff. As shown in Fig. 5, CTA-Diff demonstrates clear advantages in image quality and detail preservation. Specifically, compared to FGDM, it produces more detailed carotid and vertebral arteries, accurately capturing vascular topology and better matching the ground truth in vessel diameter (e.g., Row 3, Columns 1 and 8). Compared to CDM, CTA-Diff effectively avoids artifacts such as enlarged vessels and misaligned structures, accurately highlighting vascular regions in line with clinical expectations (e.g., Row 4, Columns 3 and 8). While achieving comparable results to TPDM, our method offers enhanced vascular detail, generating structures that more closely resemble the ground truth (e.g., Row 6, Columns 7 and 8). Overall, these results consistently demonstrate superior visual coherence and structural fidelity, underscoring the model's potential clinical value.

Relying solely on 2D cross-sectional CTA images is often insufficient for clinical diagnosis, as physicians require comprehensive 3D vascular representations for accurate assessment. Therefore, CTA-Diff must not only generate visually accurate 2D CTA images but also ensure structural consistency across sagittal and 3D views. To evaluate multi-view alignment and structural fidelity, CTA-Diff and existing methods were applied to generate full-volume CT-based CTA images of the head and neck region, followed by 3D vascular reconstruction. Specifically, 3D Slicer¹ was used for 3D visualization. The reconstructed 3D images were then compared with the ground-truth (GT) CTA vascular structures. As shown in Fig. 6, CTA-Diff achieves superior vascular continuity and structural integrity, avoiding irregular artifacts such as discontinuities and protrusions, and outperforming existing methods. For the 3D visualizations in Fig. 6, CTA-Diff is applied independently to each axial CT slice to generate slice-wise CTA predictions, which are then stacked according to the original DICOM spacing to form a full 3D volume for qualitative comparison with ground-truth CTA. These results further highlight the model's clinical applicability and its potential to enhance diagnostic accuracy.

In addition to objective metrics, a preliminary subjective evaluation of 3D visualizations was conducted by an experienced radiologist with over 10 years in vascular imaging. As shown in Table 1, for a subset of test subjects, 3D volumes reconstructed by stacking slices from real

¹ 3DSlicer: <https://www.slicer.org/>

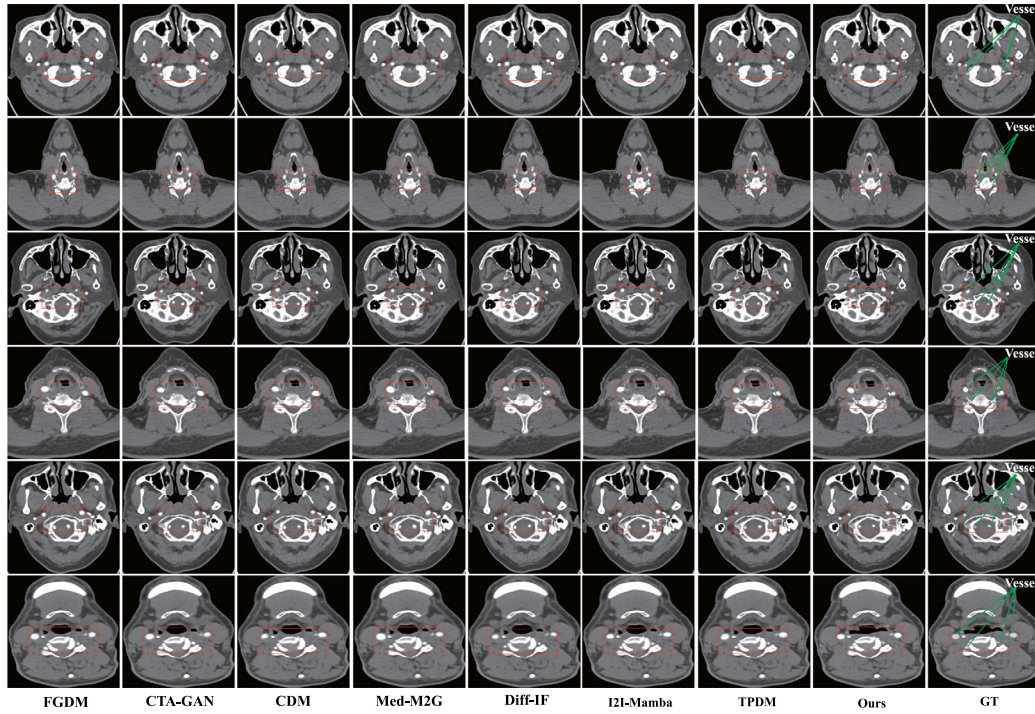


Fig. 5. Generated CTA images on the neck dataset, the red boxes indicate areas of interest. In GT, the upper two vessels are stiffer arteries, while the lower two are vertebral arteries.

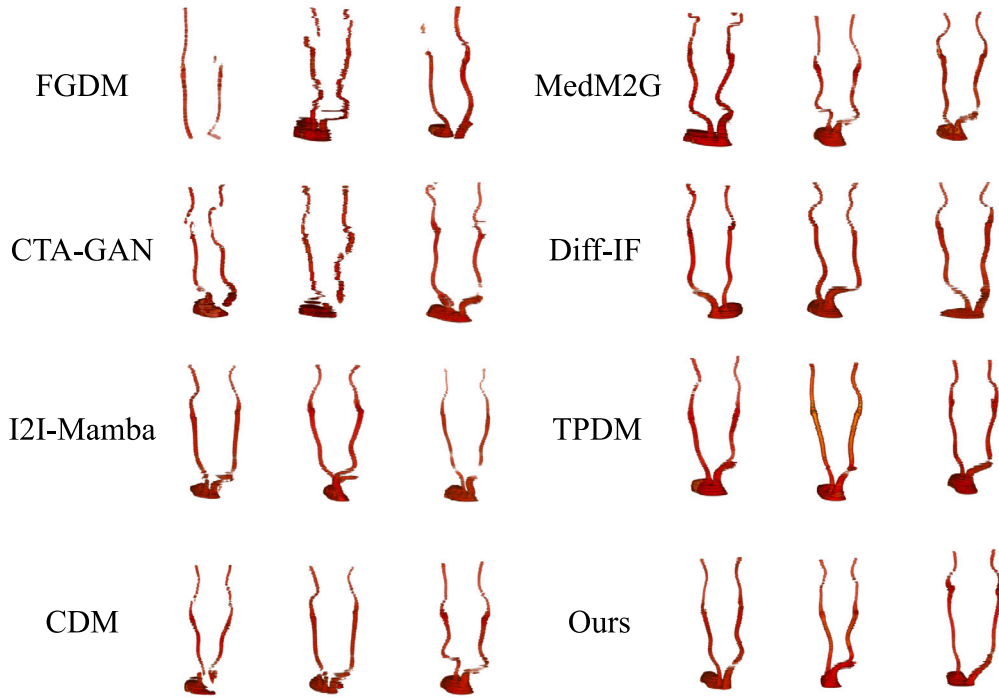


Fig. 6. 3D reconstructions of the carotid artery obtained by stacking 2D CTA slices, shown for three representative cases.

CTA and CTA-Diff were rendered in 3D Slicer and presented in random order. The radiologist rated overall image quality and vessel visibility on a 5-point scale (1 = non-diagnostic, 5 = excellent) and judged whether each reconstruction was acceptable for clinical visualization of major neck vessels. Most CTA-Diff visualizations were rated good or excellent and considered acceptable, consistent with the quantitative PSNR and SSIM improvements shown in [Table 2](#).

4.2.2. Quantitative analysis

For a more comprehensive evaluation, we conducted a quantitative assessment using PSNR, SSIM, and NMAE, systematically comparing CTA-Diff with several state-of-the-art (SOTA) models. As shown in [Table 2](#), CTA-Diff consistently outperforms all baselines across evaluation metrics, achieving an NMAE of 0.009, SSIM of 0.948, and PSNR of 38.88. Compared to 2023 methods, it markedly improves

Table 1

Radiologist subjective assessment of the reconstructed 3D images generated by our model (final output), using a 5-point scale (1 = nondiagnostic, 5 = excellent). Scores are reported as mean $\pm$ sd, and the percentage of cases rated as good/excellent (score ≥ 4) is shown.

Methods	Neck dataset		Abdomen dataset	
	Mean score	Cases ≥ 4 (%)	Mean score	Cases ≥ 4 (%)
Real CTA	4.8 $\pm$ 0.4	100	4.7 $\pm$ 0.5	100
CTA-Diff (Ours)	4.4 $\pm$ 0.4	92	4.1 $\pm$ 0.8	75

Table 2

Quantitative comparison of different methods on the neck and abdominal datasets (mean $\pm$ sd over test subjects). The best and second-best results are highlighted in red and blue, respectively. The bottom row (“ p -value (ANOVA)”) reports the global p -values from a one-way repeated-measures ANOVA across all eight models for each metric (NMAE, PSNR, SSIM) on each dataset. The “Significant vs CTA-Diff” columns list Bonferroni-adjusted p -values from post-hoc paired t -tests comparing each baseline with CTA-Diff at the patient level.

Models	Neck dataset				Abdominal dataset			
	NMAE $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	Significant vs CTA-Diff	NMAE $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	Significant vs CTA-Diff
CTA-GAN'23	0.021 $\pm$ 0.0011	25.19 $\pm$ 4.49	0.918 $\pm$ 0.08	8.421 $\times 10^{-5}$	0.023 $\pm$ 0.0010	24.96 $\pm$ 5.22	0.904 $\pm$ 0.07	6.342 $\times 10^{-4}$
FGDM'23	0.026 $\pm$ 0.0016	26.40 $\pm$ 6.83	0.908 $\pm$ 0.06	3.134 $\times 10^{-4}$	0.028 $\pm$ 0.0022	24.90 $\pm$ 8.32	0.883 $\pm$ 0.08	5.424 $\times 10^{-4}$
I2I-Mamba'24	0.018 $\pm$ 0.0024	29.18 $\pm$ 2.48	0.948 $\pm$ 0.04	9.324 $\times 10^{-5}$	0.021 $\pm$ 0.0019	28.61 $\pm$ 3.98	0.900 $\pm$ 0.07	3.914 $\times 10^{-5}$
CDM'24	0.012 $\pm$ 0.0017	33.04 $\pm$ 5.73	0.934 $\pm$ 0.05	7.934 $\times 10^{-5}$	0.015 $\pm$ 0.0016	31.12 $\pm$ 7.94	0.896 $\pm$ 0.08	8.343 $\times 10^{-5}$
MedM2G'24	0.023 $\pm$ 0.0018	32.45 $\pm$ 2.26	0.899 $\pm$ 0.02	4.394 $\times 10^{-5}$	0.026 $\pm$ 0.0017	29.89 $\pm$ 2.87	0.870 $\pm$ 0.03	5.572 $\times 10^{-5}$
Diff-IF'24	0.015 $\pm$ 0.0020	32.95 $\pm$ 4.51	0.927 $\pm$ 0.06	5.832 $\times 10^{-4}$	0.018 $\pm$ 0.0022	30.25 $\pm$ 6.75	0.906 $\pm$ 0.07	4.984 $\times 10^{-4}$
TPDM'25	0.011 $\pm$ 0.0011	38.80 $\pm$ 6.23	0.931 $\pm$ 0.06	9.234 $\times 10^{-4}$	0.013 $\pm$ 0.0012	37.89 $\pm$ 5.56	0.875 $\pm$ 0.08	6.862 $\times 10^{-4}$
CTA-Diff (Ours)	0.009 $\pm$ 0.0014	38.88 $\pm$ 6.31	0.948 $\pm$ 0.04	–	0.012 $\pm$ 0.0008	37.92 $\pm$ 8.01	0.906 $\pm$ 0.07	–
p -value (ANOVA)	1.387 $\times 10^{-7}$	4.619 $\times 10^{-8}$	2.128 $\times 10^{-7}$	–	9.823 $\times 10^{-8}$	3.285 $\times 10^{-7}$	7.512 $\times 10^{-8}$	–

image quality, reducing NMAE by 0.017 and increasing SSIM and PSNR by 0.03 and 12.48, respectively, relative to FGDM. Several 2024 models, including I2I-Mamba, MedM2G, and CDM, surpass FGDM but remain inferior to CTA-Diff; for example, MedM2G exhibits an NMAE 0.014 higher than ours, with SSIM and PSNR 0.049 and 6.43 lower, respectively. Compared with the latest 2025 method TPDM, CTA-Diff achieves lower NMAE (0.009 vs. 0.011), higher SSIM (0.948 vs. 0.931), and slightly higher PSNR (38.88 vs. 38.80), demonstrating superior performance across all three metrics.

All quantitative metrics are computed at the patient level as described above. To assess whether the choice of model has a statistically significant effect on reconstruction quality, we applied a one-way repeated-measures analysis of variance (ANOVA) to patient-level NMAE, PSNR, and SSIM across all eight methods for each dataset (neck and abdomen). Normality of the patient-level metric distributions was checked using the Shapiro–Wilk test and visual inspection of Q–Q plots, which did not reveal substantial deviations from normality; thus, the use of repeated-measures ANOVA was considered appropriate. The resulting global p -values are reported in the bottom row of Table 2 (“ p -value (ANOVA)”). In addition, post-hoc pairwise comparisons between CTA-Diff and each baseline were performed using paired t -tests with Bonferroni correction at the patient level. The Bonferroni-adjusted p -values from these post-hoc tests are listed in the “Significant vs CTA-Diff” columns for the neck and abdominal datasets, while the CTA-Diff rows contain “–” because no comparison is defined for the reference method.

4.3. Ablation analysis

To evaluate the effectiveness of CTA-Diff, we conducted a systematic ablation study by progressively removing or modifying key components of the generation model. Three ablation scenarios were designed for the CT-to-CTA task, with results summarized in Table 3. In our model, E_{c1} encodes CT image content, E_{c2} encodes CTA image content, and E_s extracts CTA style features. Models M_1 through M_6 correspond to different combinations of these modules, allowing us to assess their individual contributions as well as their joint effects on model performance. The results show that removing any single component leads to noticeable performance drops. Among all variants, M_1 performed the worst (NMAE: 0.042; SSIM: 0.900), significantly lower than the

full model (NMAE: 0.009; SSIM: 0.948). In contrast, M_5 performed best among the ablations, with only slight degradation (NMAE: 0.012; SSIM: 0.926). These findings highlight the critical roles of E_{c2} and E_s . Among the models retaining only one encoder (M_2 – M_4), M_2 (with only E_{c1}) showed the weakest performance. Notably, M_5 outperformed M_6 , indicating that excluding E_{c1} had less impact than excluding E_{c2} or E_s . Overall, the analysis confirms that E_{c2} and E_s are key to achieving optimal CT-to-CTA synthesis performance. As shown in Fig. 7, M_1 shows the poorest generation quality, with pronounced discrepancies from real CTA images. Among M_2 – M_6 , M_2 —which lacks both E_{c2} and E_s —performs suboptimally, generating only vague vascular structures. In contrast, M_5 , omitting only E_{c1} , achieves the best results among the ablated models. It successfully reconstructs key arteries, particularly the carotids, though its output still falls short of the complete model in overall fidelity.

We further contrast the impact of different downsampling strategies in the diffusion Mamba-UNet backbone. In the pooling variant, all patch merging layers are replaced with standard average pooling with a stride of 2, while the rest of the architecture and training protocol remain unchanged. As summarized in Table 4, this variant yields consistently lower PSNR, SSIM, and NMAE on both the neck and abdominal cohorts, indicating that the learnable patch merging better preserves and aggregates local token information than fixed pooling, thereby improving generation quality.

4.4. Performance analysis about proposed method

4.4.1. Generalization performance on the abdominal dataset

To evaluate the generalization capability of CTA-Diff across anatomical regions, we applied it to the abdominal CT-to-CTA generation task and compared its performance with existing abdominal CTA generation methods. The generated images are shown in Fig. 8. Compared to the neck CTA task, performance metrics for the abdominal task show a decline, reflecting the increased anatomical complexity and dense vascular structures in the mid-abdominal region. Despite this challenge, CTA-Diff consistently achieves superior image quality. As summarized in Table 2, it attains an NMAE of 0.012, outperforming FGDM by 0.016. The PSNR reaches 37.92, only 0.97 below the best result of 38.89, while SSIM remains high at 0.906, highlighting the model’s robustness. These results demonstrate that CTA-Diff generalizes effectively to CT-to-CTA synthesis beyond the neck region.

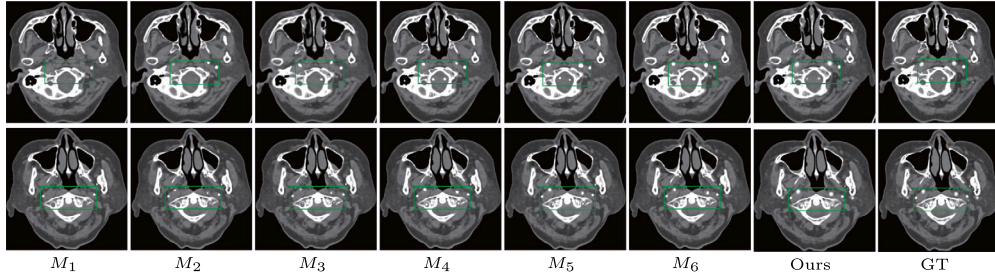


Fig. 7. Ablation study: Qualitative comparison of different models on the neck dataset. The green boxes indicate areas of interest, including the carotid and vertebral arteries. M_1 removes all three modules (no E_{c1} , E_{c2} , or E_s). M_2 , M_3 , and M_4 retain only E_{c1} , only E_{c2} , and only E_s , respectively. M_5 retains E_{c2} and E_s , M_6 retains E_{c1} and E_s , and M_7 corresponds to the full CTA-Diff model with all three modules enabled.

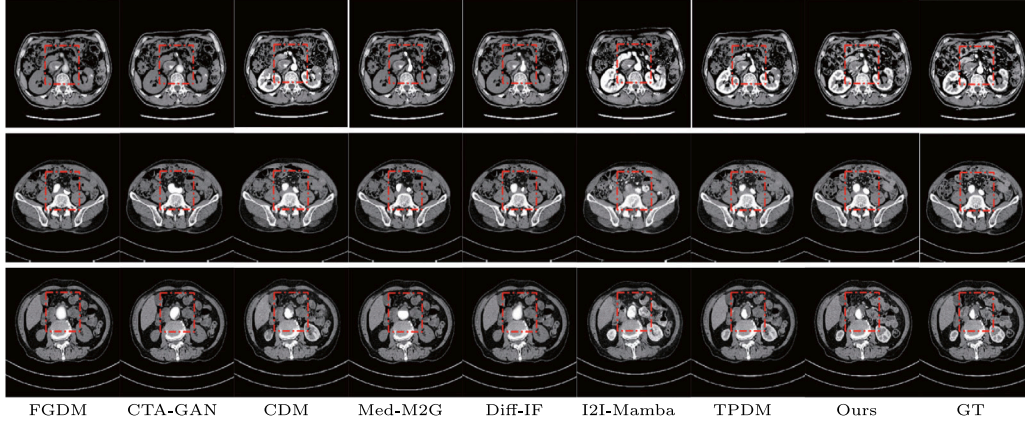


Fig. 8. Generated CTA images on the abdominal dataset, the red boxes indicate areas of interest, including the abdominal aorta formation.

Table 3

Ablation study evaluating different model variants. E_{c1} , E_{c2} , and E_s denote the CT content encoder, CTA content encoder, and CTA style encoder, respectively. M_1 removes all three modules (no E_{c1} , E_{c2} , or E_s). M_2 , M_3 , and M_4 retain only E_{c1} , only E_{c2} , and only E_s , respectively. M_5 retains E_{c2} and E_s , M_6 retains E_{c1} and E_s , and M_7 corresponds to the full CTA-Diff model with all three modules enabled.

Models	E_{c1}	E_{c2}	E_s	NMAE ↓	SSIM ↑	PSNR ↑
M_1	×	×	×	0.042	0.900	36.23
M_2	✓	×	×	0.039	0.902	36.39
M_3	×	✓	×	0.031	0.904	36.50
M_4	×	×	✓	0.024	0.909	36.81
M_5	×	✓	✓	0.012	0.926	37.84
M_6	✓	×	✓	0.019	0.913	36.85
M_7 (Ours)	✓	✓	✓	0.009	0.948	38.88

Table 4

Comparison of pooling and patch merging strategies in the diffusion Mamba-UNet backbone.

Methods	Neck dataset			Abdomen dataset		
	NMAE ↓	PSNR ↑	SSIM ↑	NMAE ↓	PSNR ↑	SSIM ↑
Pooling-based strategy	0.030 ± 0.0015	20.10 ± 5.98	0.743 ± 0.05	0.035 ± 0.0011	19.30 ± 7.12	0.699 ± 0.07
Merging-based strategy (Ours)	0.009 ± 0.0014	38.88 ± 6.31	0.948 ± 0.04	0.012 ± 0.0008	37.92 ± 8.01	0.906 ± 0.07

4.4.2. Robustness of CTA-Diff against Gaussian noise

To systematically assess the impact of noise on model performance, Gaussian noise with zero mean and varying variances was added to all test images. The model was then evaluated on these noise-affected datasets. As shown in Fig. 9, performance gradually declines with increasing noise. At a variance of 0.02, CTA-Diff remains stable with minor fluctuations, demonstrating strong robustness under low-noise conditions. Performance drops significantly at a variance of 0.1, highlighting the importance of considering noise effects in high-noise clinical settings. Overall, CTA-Diff maintains satisfactory CT-to-CTA generation quality for low to moderate noise levels but deteriorates under heavy noise.

4.4.3. Model complexity analysis

This section compares CTA-Diff with existing methods in terms of parameter count and model performance. Despite its relatively high parameter count, CTA-Diff achieves substantially improved performance. As shown in Fig. 10, it consistently outperforms competing methods across key metrics. Specifically, with over 50M parameters – the second highest among the evaluated models – CTA-Diff attains a PSNR of 38.88 on the cervical dataset, surpassing Diff-IF (23.74M parameters) by 6.29. It also achieves the highest SSIM and NMAE values of 0.948 and 0.009, respectively.

In addition to parameter counts, we evaluated the inference efficiency of CTA-Diff. Using the default sampling configuration and an

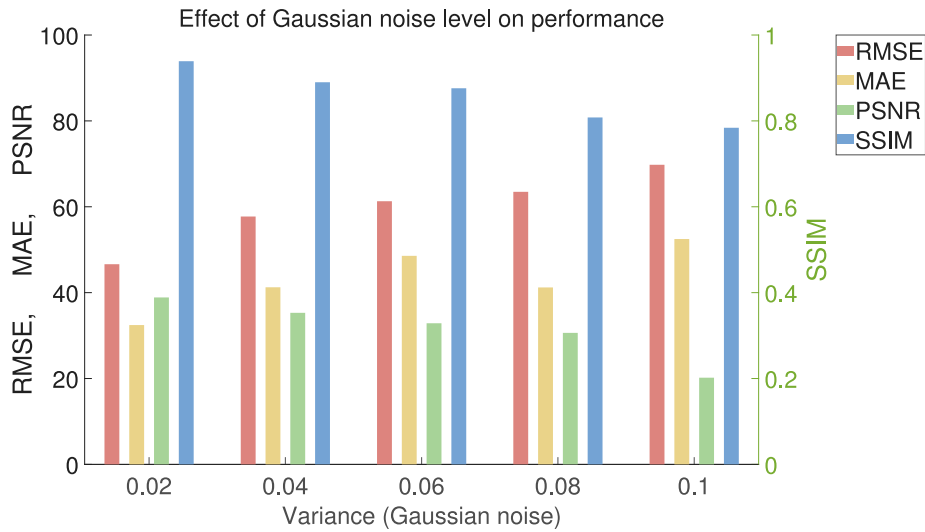


Fig. 9. Performance of CTA-Diff under zero-mean Gaussian noise with varying variance.

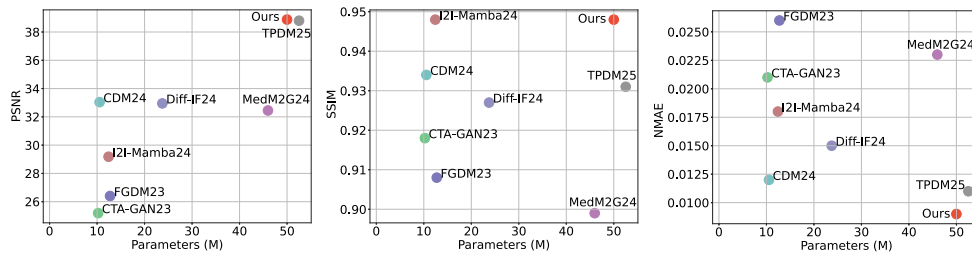


Fig. 10. The number of parameters and corresponding performance of different methods on the neck dataset.

input resolution of 512×512 , the average inference time on an NVIDIA GeForce RTX 4090D is approximately 40 ms per slice, corresponding to about 18 s per full neck volume ($\approx 300\text{--}400$ slices). On a more modest GPU (NVIDIA 3060), the corresponding times are 120 ms per slice and 42 s per full volume. These results indicate that CTA-Diff can synthesize clinically usable CTA-like volumes within a time frame compatible with routine radiological workflows, especially on modern high-end GPUs, while remaining feasible on mid-range hardware.

4.5. Failure cases analysis

While CTA-Diff generates high-quality CTA images in most cases, it still faces challenges in highly demanding scenarios. As shown in Fig. 11, the first several columns depict poor predictions, whereas the last two columns serve as examples of successful generation. Failure cases (e.g., columns 1 and 6) often arise from substantial intensity discrepancies between corresponding CT and CTA regions, causing the model to overemphasize highly contrasted areas and lose critical vascular details. Additionally, the stochastic nature of the diffusion process can introduce spatial discontinuities, particularly as inconsistencies between coronal and sagittal views. Extracting reliable anatomical structures from CT images also remains difficult; although the proposed CT content encoder generally performs well, it may struggle under such complex conditions.

4.6. Limitations and discussion

First, while CTA-Diff achieves superior image-quality metrics and produces visually realistic vessel appearances, these results do not necessarily guarantee improved diagnostic efficacy in clinical practice. The generated images are intended as a supportive tool rather than

a replacement for real CTA, and all clinical decisions must involve radiologist oversight.

Second, the clinical utility of CTA-Diff has not been established. In real clinical settings, radiologists must detect vascular stenosis or occlusion and estimate stenosis severity. However, the retrospective datasets used in this study lack systematic lesion-level annotations or reference measurements, so diagnostic performance metrics – such as sensitivity, specificity, and reader-based accuracy – were not evaluated. Future studies will focus on prospective evaluation in patients with confirmed vascular pathologies to assess the true diagnostic efficacy of CTA-Diff.

Third, a limitation of CTA-Diff is that it is currently optimized using pixel-level reconstruction and global content/style losses, without explicit lesion-aware or task-specific constraints. As a result, subtle focal abnormalities, such as mild stenoses or small plaques, may not be optimally preserved, even when global image-quality metrics remain high. This could be addressed in future work by incorporating region-weighted or vessel-aware loss terms near annotated lesions and along the vascular tree.

Fourth, the current diffusion Mamba-UNet backbone is trained purely in an image-to-image fashion and lacks auxiliary heads for stenosis detection, vessel grading, or centerline-lumen analysis, which could provide additional supervision around clinically critical regions. A natural extension would be to adopt a multi-task architecture with segmentation or classification branches, allowing reconstruction and lesion understanding to be learned jointly. Together with lesion-aware training, this could further improve the preservation and accurate representation of clinically relevant features.

5. Conclusions and future work

This paper presents CTA-Diff, a generative model for converting CT images into CTA images, reducing clinical reliance on ICA for CTA

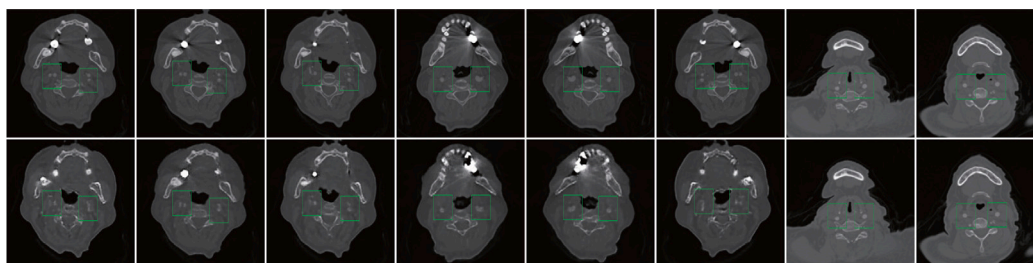


Fig. 11. Examples of CTA images predicted by CTA-Diff for the carotid artery. Ground-truth CTA images are shown in the first row, and the corresponding predictions are shown in the second row. The first six columns illustrate failure cases, whereas the last two columns show successful predictions. Green boxes indicate the carotid artery regions, highlighting areas with suboptimal or successful generation quality.

imaging. Unlike GAN-based methods, CTA-Diff leverages a diffusion model built on the Mamba-UNet architecture for efficient CT-to-CTA generation. To enhance feature extraction, modality-specific content conditioning for CT and CTA images ensures structural consistency in the generated CTA images. Additionally, CTA-specific style conditioning and a style loss function are used to align the generated images with real CTA styles. Extensive experiments validate the model's effectiveness, achieving superior performance on neck and abdominal datasets, with PSNR, SSIM, and NMAE scores of 38.88, 0.948, and 0.009 for the neck dataset and 37.92, 0.906, and 0.012 for the abdominal dataset.

Future work will proceed along two main directions. First, we plan to further exploit the frequency-domain characteristics of CT images by exploring alternative transformations, such as the discrete wavelet transform (DWT). Given DWT's strength in capturing local frequency and structural granularity, we aim to construct an optimized anatomical enhancement map that highlights subtle CT features and enhances the network's representational capacity. Second, to improve model performance under challenging conditions – particularly with large domain or style discrepancies such as highly contrasted regions – we propose replacing the traditional Gram matrix with the contrastive arbitrary style transfer (CAST) framework (Zhang et al., 2022b). By integrating both CT and CTA modalities within a contrastive learning scheme, the model can learn from positive and negative style representations, thereby enhancing adaptability, robustness, and generalization in complex clinical scenarios.

CRediT authorship contribution statement

Yuhang Xia: Writing – original draft, Software, Methodology. **Hongqing Zhu:** Supervision, Project administration, Funding acquisition. **Tong Hou:** Visualization, Validation. **Ning Chen:** Investigation, Formal analysis. **Kai Chen:** Resources, Data curation. **Zhong Zheng:** Visualization, Resources. **Bingcang Huang:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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